

MATH3171 (Mathematical Biology III)
YEAR 2025–26, MICHAELMAS TERM

PROBLEM SHEET 2

This problem sheet is due in by **5pm** on **Wednesday 12 November 2025**. You may submit it on paper to my office (MCS 3008) or electronically on Gradescope. There is a discretionary mark available for neat and legible solutions.

Topics covered: Chapter 3.

Consider this variation on competitive Lotka–Volterra, where the interaction terms are of a predator–prey type:

$$\frac{dx}{dt} = ax \left(1 - \frac{x}{\eta_1}\right) - bxy, \quad (1a)$$

$$\frac{dy}{dt} = cy \left(1 - \frac{y}{\eta_2}\right) + dxy, \quad (1b)$$

where $a, b, c, d, \eta_1, \eta_2 > 0$.

1. (a) Give a short biological interpretation of the parameters $a, b, c, d, \eta_1, \eta_2$.
(b) Which of x and y might represent a predator, and which might represent prey?
(c) In traditional Lotka–Volterra, the predator dies out if the prey population is zero. Is that the case here? What does this say about the predator’s diet in this model?
2. By setting $x = \hat{x}X$, $y = \hat{y}Y$, $t = \hat{t}T$, nondimensionalise the system and then remove hats to obtain

$$\frac{dx}{dt} = x(1 - x - \gamma_1 y), \quad (2a)$$

$$\frac{dy}{dt} = \beta y(1 - y + \gamma_2 x), \quad (2b)$$

where $\beta, \gamma_1, \gamma_2 > 0$. In doing so, give the definition of the new parameters in terms of the old parameters.

3. By sketching the nullclines of the system for two different cases, show how the number of permissible equilibria depends on γ_1 . (Note this is different to the competitive Lotka–Volterra model we saw in class, where the number of permissible equilibria depended on both γ_1 and γ_2 .)
4. The Jacobian of the linearised system, $\mathbf{J}$, is

$$\mathbf{J} = \begin{pmatrix} 1 - 2x_0 - \gamma_1 y_0 & -\gamma_1 x_0 \\ \beta \gamma_2 y_0 & \beta(1 - 2y_0 + \gamma_2 x_0) \end{pmatrix}, \quad (3)$$

and, similarly to competitive Lotka–Volterra, there are four equilibria, at

$$(x_0, y_0) = (0, 0), \quad (1, 0), \quad (0, 1) \quad \text{and} \quad \left(\frac{1 - \gamma_1}{1 + \gamma_1 \gamma_2}, \frac{1 + \gamma_2}{1 + \gamma_1 \gamma_2} \right).$$

The characteristic equation for this final equilibrium is

$$\lambda^2(1 + \gamma_1\gamma_2) + \lambda(1 - \gamma_1 + \beta[1 + \gamma_2]) + \beta(1 - \gamma_1)(1 + \gamma_2) = 0. \quad (4)$$

Recalling that $\beta, \gamma_1, \gamma_2 > 0$, show that for any given parameter set $(\beta, \gamma_1, \gamma_2)$, all equilibria are unstable *apart from exactly one*.

Hint: It will help to fill this table with ‘stable’, ‘degenerate’ or ‘unstable’. Remember that a saddle point is a type of unstable equilibrium (because it’s unstable in at least one direction).

	$(0, 0)$	$(0, 1)$	$(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$	$(1, 0)$
$\gamma_1 < 1$	?	?	?	?
$\gamma_1 = 1$	?	?	?	?
$\gamma_1 > 1$	?	?	?	?

- For the case $\gamma_1 < 1$, draw a complete phase portrait, including some sample trajectories, and confirm your findings with regards to the stability of the equilibria for this case in question 4. You should be able to start by using one of your nullcline sketches from question 3.
- Finally, in light of the results in question 4, discuss the validity of the model for a biological system. As part of this analysis, consider the physical interpretation of each equilibrium and whether it would be desirable for it to be stable or not.